

TrES-1b

R-1428

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_200930 · created 2026-08-01T07:08:06+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459122.7098 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.022
Transit depth (Rp/Rs)^2	3.37 +/- 0.82 [%]
Orbital Inclination (inc)	89.13 +/- 1.17
Airmass coefficient 1 (a1)	1.37 +/- 0.022
Airmass coefficient 2 (a2)	-0.163 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.52 %
Best Comparison Star	None
Optimal Aperture	4.16
Optimal Annulus	18.44
Transit Duration (day)	0.096 +/- 0.0031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.022 vs 0.1358 (deviation 2.19 σ)
Duration fit vs published	0.096 d vs 0.1042 d (deviation 2.63 σ)
Mid-time O-C	-6.0 min
Depth SNR	8.4
Residual scatter	1.52 %

Light curve

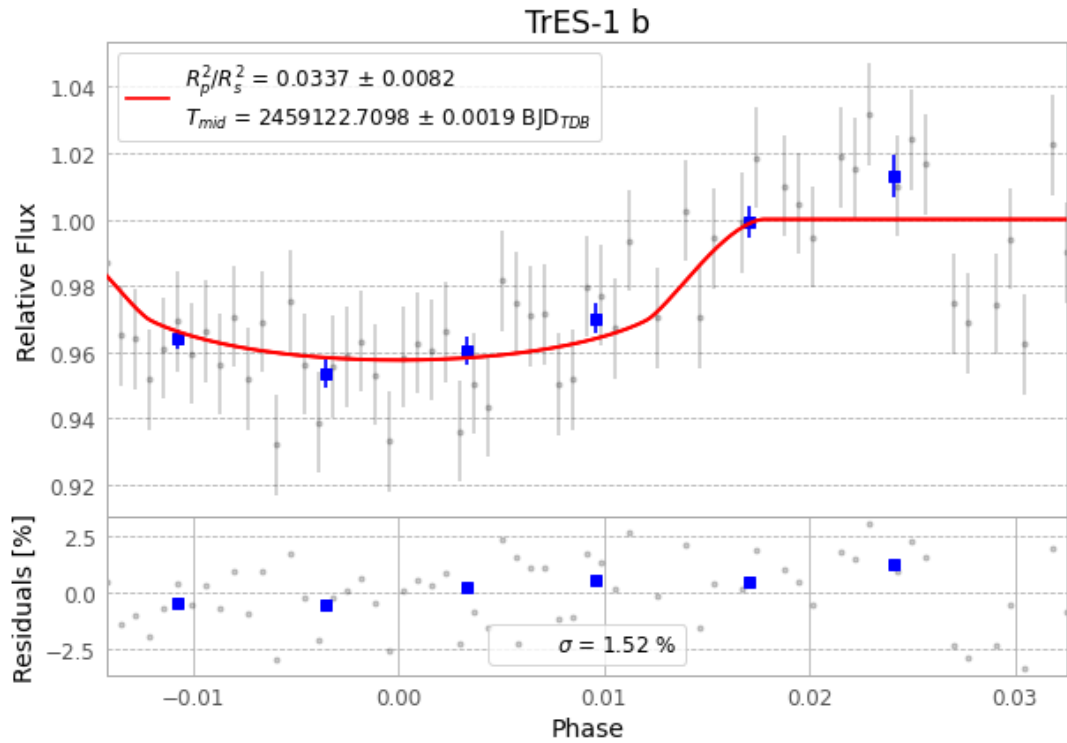


Figure 1 — FinalLightCurve_TrES-1 b_2020-09-29.png (SHA-256 ce2b17015cdd77c0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [262, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 203e55d8ed30bfb8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

60 FITS frames (39 MB), TRES-1200930035713.FITS → TRES-1200930072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-200930.log (df9d82cff55c...), FinalLightCurve_TrES-1 b_2020-09-29.png (ce2b17015cdd...), FinalLightCurve_TrES-1 b_2020-09-29.csv (835a2901bcfa...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:08Z.